

ASSIGNMENT

FoC

Anjali Arun
CSE-A
Roll no 25

①
• Write the actions performed for recovering a file when you accidentally deleted it in linux environment.

When you delete a file in linux using `rm`, it doesn't go to a trash bin — it's unlinked from the filesystem. However, recovery is often possible if you act quickly.

Step 0: Stop writing to the disk immediately

Step 1: Check if the file is still open by a process

```
bash
```

```
# List open file descriptors
```

```
ls -l | grep deleted
```

```
# if found, note the PID & FD no, then copy the back
```

```
cp /proc/<PID>/fd/<FD>/path/to/recovered-file
```

Step 2: Check the trash (desktop environment)

If you deleted the file via a file manager (Nautilus, Dolphin, etc.) it may be in the trash.

```
bash
```

```
ls -l ~/.local/share/Trash/files/
```

```
# Restore it
```

```
mv ~/.local/share/Trash/files/filename/path/to/restore/
```

Step 3: Restore from backup

Check if a backup exists

```
bash
```

```
# If using rsync/timeshift/tar backups
```

```
ls /backup/location/
```

Timeshift (system snapshots)

```
sudo timeshift --list
```

```
sudo timeshift --restore
```

Step 4: Use extundelete (for ext3/4 filesystems)

```
bash
```

Install

```
sudo apt install extundelete
```

Unmount the partition first (use a live USB if it's the root partition)

```
sudo umount /dev/sda1
```

Recover a specific file

```
sudo extundelete /dev/sda1 --restore -file path/to/deleted/
```

Recover all deleted files

file

```
sudo extundelete /dev/sda1 --restore1 -all
```

Recovered files appear in ./RECOVERED-FILES/

Step 5: testdisk / photorec

testdisk is a powerful recovery tool for partitions & files

```
bash
```

Install

```
sudo apt install testdisk
```

Launch

```
sudo testdisk /dev/sda
```

Navigate : Analyse → Quick search → Advanced → Undelete

Then select & copy the file to a safe location

(2)

photorec (bundled with testdisk) recovers files by type

bash

```
sudo photorec /dev/sda1
```

Follow the interactive prompts to select partition & file types

Step 6: Use foremost or scalpel (carving tool)
These tools ~~from~~ scan raw disk data for file signature:

bash

```
# Install foremost
```

```
sudo apt install foremost
```

```
# Recover to from a disk image or partition
```

```
sudo foremost -i /dev/sda1 -o /recovery/output/
```

```
# Install scalpel
```

```
sudo apt install scalpel
```

```
sudo scalpel /dev/sda1 -o /recovery/output/
```

Step 7: Create a disk image first (Best practice before recovery)

Before running any recovery tool, image the disk to avoid further damage.

bash

```
sudo dd if=/dev/sda1 of=/external/drive/disk-image.img  
bs=4096 conv=noerror,sync
```

```
# Then run recovery tools on the image, not the live disk
```

```
sudo extundelete /external/drive/disk-image.img --restore-all
```